Sahand Rezaei-Shoshtari

PhD Candidate in Computer Science

sahandrez.github.io | github.com/sahandrez | linkedin.com/in/sahand-rezaei

Education

Present – Sep. 2020 | **PhD, School of Computer Science, McGill University and Mila, Montreal, Canada**
Supervisors: Doina Precup, David Meger
CGPA: 4.00/4.00

Dec. 2019 – Sep. 2017 | **Master of Engineering - Thesis, McGill University, Montreal, Canada**
Supervisors: Inna Sharf, David Meger
CGPA: 4.00/4.00
Thesis: Learning Manipulator Dynamics for Control and Interaction Inference

Sep. 2016 – Sep. 2012 | **Bachelor of Mechanical Engineering, University of Tehran, Tehran, Iran**
CGPA: 3.98/4.00
Thesis: Online Path Planning for a Mobile Robot in Dynamic Environments using Reinforcement Learning

Work Experience

Dec. 2024 – Apr. 2024 | **Researcher, Microsoft Research AI for Science, Montreal, Canada**
- Reinforcement learning and diffusion models for material discovery

Aug. 2023 – May. 2023 | **Research Intern, Microsoft Research AI for Science, Amsterdam, The Netherlands**
- Reinforcement learning and diffusion models for material discovery

Feb. 2023 – Apr. 2022 | **Research Intern, Samsung AI Center, Montreal, Canada**
- Reinforcement learning and meta learning with transformers and hypernetworks

Sep. 2020 – Mar. 2020 | **Research Intern, Samsung AI Center, Montreal, Canada**
- Multimodal generative modeling for learning dynamical models
- Reinforcement learning for load balancing of 5G networks

Mar. 2020 – Jan. 2020 | **AI Programmer, Ubisoft La Forge, Montreal, Canada**
- Reinforcement learning for automated video game testing

Aug. 2019 – Mar. 2019 | **Research Intern, Samsung AI Center, Montreal, Canada**
- Computer vision and convolutional neural networks (CNNs) for human-robot interaction

Publications

2024 **Rezaei-Shoshtari, S.**, Yurchyk, H., Fujimoto, S., Precup, D., and Meger, D., 2024. "Fairness in Reinforcement Learning with Bisimulation Metrics". *Pre-print*.

2024 Panangaden, P.*, **Rezaei-Shoshtari, S.***, Zhao, R.*, Meger, D., and Precup, D., 2024. "Policy Gradient Methods in the Presence of Symmetries and State Abstractions". *Journal of Machine Learning Research (JMLR)*.

2023 **Rezaei-Shoshtari, S.**, Morissette, C., Hogan, F.R., Dudek, G., and Meger, D., 2023. "Hypernetworks for Zero-shot Transfer in Reinforcement Learning". *The AAAI Conference on Artificial Intelligence*.

2022 **Rezaei-Shoshtari, S.**, Zhao, R., Panangaden, P., Meger, D., and Precup, D., 2022. "Continuous MDP Homomorphisms and Homomorphic Policy Gradient". *Advances in Neural Information Processing Systems (NeurIPS)*.

2021 **Rezaei-Shoshtari, S.**, Hogan, F.R., Jenkin, M., Meger, D., and Dudek, G., 2021. "Learning Intuitive Physics with Multimodal Generative Models". *The AAAI Conference on Artificial Intelligence*.

2021 Hogan, F.R., Jenkin, M., **Rezaei-Shoshtari, S.**, Girdhar, Y., Meger, D., and Dudek, G., 2021. "Seeing Through your Skin: Recognizing Objects with a Novel Visuotactile Sensor". *The IEEE/CVF Winter Conference on Applications of Computer Vision*.

2020 Molamohammadi, M., **Rezaei-Shoshtari, S.**, and Quitoriano, N., 2020. "Jacobian of generative models for sensitivity analysis of photovoltaic device processes". *The Machine Learning for Engineering Workshop at NeurIPS*.

2020 **Rezaei-Shoshtari, S.**, Meger, D., and Sharf, I., 2020. "Learning the Latent Space of Robot Dynamics for Cutting Interaction Inference". *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

2019 **Rezaei-Shoshtari, S.**, Meger, D., and Sharf, I., 2019. "Cascaded gaussian processes for data-efficient robot dynamics learning". *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Skills

Programming	Python, C++, C#, MATLAB
Machine Learning Frameworks	PyTorch, TensorFlow, JAX, Scikit-Learn
Large Language Models	Hugging Face Transformers/Datasets/TRL/Accelerate, vLLM, DeepSpeed, TensorRT
Platforms	Docker, Slurm Workload Manager, Azure
Other Software	Git, Unity 3D, Weights & Biases, OpenCV, Mujoco, ROS, Bullet

Select Projects

LLM Finetuning with Reinforcement Learning from Unpaired Human Feedback — 2024

github.com/sahandrez/unpaired_rlhf

An approach for finetuning LLMs with reinforcement learning using unpaired preference data (thumbs up/down preferences).

RLHF | LLM | PyTorch | Hugging Face Libraries | Transformers | TRL | Python

Hypernetworks for Zero-Shot Transfer in Reinforcement Learning — 2023

Paper Webpage github.com/sahandrez/hyperzero

A novel algorithm based on hypernetworks for zero-shot transfer in deep reinforcement learning.

RL | Hypernetworks | Meta Learning | PyTorch | Python

Contextual Control Suite — 2023

github.com/sahandrez/contextual-control-suite

A series of contextual Markov decision processes (MDP) based on Mujoco for continuous control tasks.

RL | Control | Dynamical Systems | Mujoco | Python

Deep Homomorphic Policy Gradient (DHPG) Algorithm — 2022

Paper Webpage github.com/sahandrez/homomorphic_policy_gradient

A novel algorithm for state-action abstraction and representation learning in reinforcement learning.

RL | Control | Representation Learning | PyTorch | Python

Multimodal Generative Modeling and Visuotactile Simulation — 2020

Paper Webpage github.com/sahandrez/multimodal-dynamics

A novel algorithm for multimodal generative modeling of dynamical sytems.

Generative Modeling | Multimodal | Dynamical Systems | PyTorch | Bullet | Python

Control and Motion Planning Utilities for Kinova Jaco 2 Robot — 2019

github.com/sahandrez/jaco_control

A ROS package for the Kinova Jaco 2 robot with various control and motion planning utilities.

Robotics | ROS | Python

Select Honors and Awards

2022 – 2025	Canada Graduate Scholarship-Doctoral (CGS-D). Amount of **$105,000**. NSERC.
2022 – 2026	Fonds de Recherche du Quebec - Nature et Technologies (FRQ-NT). Amount of **$70,000**. FRQ.
2022	Grad Excellence Award. Amount of **$7,000** per year. McGill University.
2022	AAAI Student Scholarship Program. AAAI 2023.
2020	DeepMind Grad Award. Amount of **$25,000**. DeepMind and McGill University.
2019	IROS Student and Developing Countries (SDC) Travel Award. IEEE/RSJ IROS 2019.
2017	Grad Excellence Award. Amount of **$5,000**. McGill University.
2012	National University Entrance Exam. Ranked **19th**. Iran.

Community Involvement

2024 – Present	Senior reviewer at RLC.
2022 – Present	Reviewer at NeurIPS, ICML, ICLR.
2022	NeurIPS 2022 Outstanding Reviewer (Top 8%). NeurIPS.
2022	ICML 2022 Outstanding Reviewer (Top 10%). ICML.

Certifications

Apr. 2023	**Trustworthy and Responsible AI Learning (TRAIL)**, Mila, Montreal, Canada
Oct. 2020	**Simons Institute Mathematics of Online Decision Making Workshop**, Virtual
Aug. 2019	**CIFAR Deep Learning and Reinforcement Learning (DLRL) Summer School**, Edmonton, Canada